How should the performance of different computer systems be compared?

clock rate?

- but this is just one of the three factors in our formula for CPU execution time!

MIPS? ("millions of instructions per second")

- MIPS = (clock rate in GHz) x 1000 / CPI
- but this reflects just two of the three factors
- program dependent

MFlops? ("millions of floating point operations per second"), or more appropriately, given increases in performance that have occurred, GFlops, TFlops, PFlops, or EFlops !!!

- notion of an "operation" more architecture independent than that of a machine language instruction
- like MIPS, program dependent

El Capitan, a supercomputer installed at the Lawrence Livermore National Laboratory in the US, is currently #1 on the “top 500” list of the fastest publicly-announced computers, and has achieved a processing rate of over 1.8 EFlops (1.8 quintillion floating point operations per second) on the “High Performance Linpack” benchmark

- configuration includes 11,340,000 processing cores in total
- power consumption of about 30 megawatts

benchmarks: programs written in a high-level language chosen to serve as the basis of performance comparison between systems

- synthetic
- kernels / simple routines (e.g. Linpack)
- real applications (e.g. as standardized by SPEC – “Standard Performance Evaluation Corporation”)

how to summarize results from running a suite of benchmarks?

SPEC uses the geometric mean, with each execution time measured relative to that on a (standardized) reference machine

- e.g. suppose have 3 benchmarks, with execution times on some system A of 3, 2, and 9, respectively

- suppose further that on the standard reference machine, these execution times were 10, 8, and 20, respectively

- then the SPEC rating for A would be the cube root of (10/3) x (8/2) x (20/9)

- in general, with N benchmarks, would take the N'th root of the product of the N relative execution time factors

Power

(reference: text Section 1.7)

clock rates no longer increasing, owing to power consumption considerations

with current technology, "dynamic" power consumption is proportional to V^2 x f, where V is the voltage and f is the transistor switching frequency

- f is proportional to clock rate; also often lower V can be used when clock rate is lower

power wall ... can't keep increasing clock rate owing to problem of cooling the processor chip!

also have "static" power consumption resulting from "leakage current" that flows even when transistors are not being switched how to reduce power consumption?

- use simpler processor designs with fewer transistors
- cut off clocking signal to components that aren't currently in use, eliminating dynamic power consumption by those components
- cut off power to components that aren't currently in use, eliminating both dynamic and static power consumption by those components
- system sleep states

in light of the "power wall", how have we continued to make dramatic improvements in performance?

most prominently, *specialization* of processing components for specific computations or types of computations (e.g., specialized cores & accelerators in smartphones, GPUs, Google's Tensor Processing Unit (TPU), …)